LOOK AFTER

G-old

Home and Social Care
for the Aging Communities

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ABSTRACT

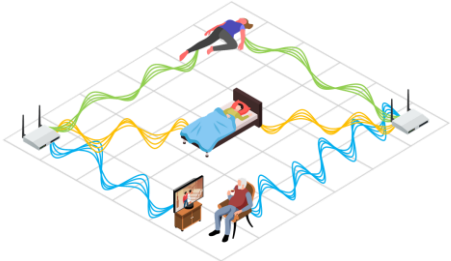
The number of people aged 65 years or older worldwide is projected to triple by 2050, reaching 1.5 billion. This demographic shift indicates that we will face an increasing demand for inexpensive and accessible home care services for the elderly. However, there are still many households with low incomes that struggle to afford specialized home care. The difficulty of providing inexpensive and accessible home care is especially severe for persons with chronic illnesses or disabilities who require continuing and specialized care to maintain their health and well-being. Among elderly individuals, respiratory diseases such as COPD(Chronic obstructive pulmonary disease) are particularly common and can have a significant impact on their quality of life. Medical researchers estimate that between 11% and 18% of adults aged 65 years or older are living with COPD. To address this, we propose "Look After", a smart home caregiver system based on Wifi sensing technology. This system is designed to tackle respiratory diseases such as COPD and asthma in elderly individuals while providing a range of other smart features to support their health and well-being. The system utilizes wifi sensing technology to detect respiratory symptoms and monitor vital signs such as respiratory rate, heartbeat rate, and sleep tracking data in real-time, allowing for early detection and intervention.

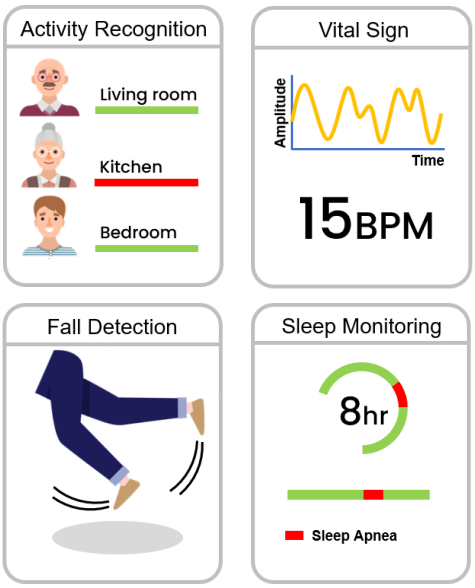
OUR SOLUTION

Advanced Feature

Respiratory Rate Monitoring:

- Specified data analysis for COPD and asthma patients.
- Respiratory rate data are collected as a time series of numeric values representing the duration of each breath in milliseconds or seconds.







Main Features

Fall detection & activity recognition:

- WiFi access points can differentiate between moving and stationary individuals.
- Motion of target can be classified into different activities, such as walking, sitting, standing, or falling.

Vital Signs Monitoring:

- Changes in chest movements can be used to estimate heart rate and respiratory rate.
- Continuous heart rate monitoring can help detect abnormal heart rhythms and other cardiovascular conditions early.
- Integrated with machine learning algorithms to detect specific

Sleep Monitoring:

- It can measure an individual's sleep patterns, total amount of sleep, and breathing during sleep.
- Estimate sleep stages and sleep disturbances to assess sleep quality and patterns.
- Useful for individuals with sleep disorders such as sleep apnea, insomnia, and restless leg syndrome.

WIFI SENSING FOR RESPIRATORY RATE MONITORING

WiFi sensing technology can monitor changes in chest size by analyzing the Channel State Information (CSI), which is information about wireless communication between a transmitter and receiver. A lot of death cases could have been reduced if the patients had more knowledge of Chronic Obstructive Pulmonary Disease (COPD). There was mislabeling of COPD as asthma by both patients and physicians. This could have resulted in a lack of understanding of treatment options, outcomes, and prognosis of COPD Patients with these conditions need to be monitored for short-term and long-term complications of COPD and for comorbidities. By tracking these changes, WiFi sensing can provide detailed respiratory data and potentially distinguish between normal breathing patterns, COPD, and asthma patterns.

For instance, there are four respiratory patterns of asthma patients detected by oronasal airflow developed for identifying irregular breathing activities when the patient has an asthma attack (Figure 1). In COPD patients, normal breathing patterns show synchronized upward deflections during inspiration and downward deflections during expiration in spirometer, chest, and abdomen signals (Figure 2). Asynchronous breathing patterns, on the other hand, display a lack of unison in chest and abdomen signals (Figure 3). The most common pattern in this category shows a synchronous chest signal with airflow, but the abdomen signal moves abruptly downward near or at the end of inspiration and upward in the expiration, continuing to move upward during some part of expiration.

It is achievable to identify and distinguish between breathing patterns associated with COPD and asthma by utilising the capabilities of WiFi sensing technology, providing a non-invasive technique for monitoring these respiratory disorders.

DATA PROCESSING METHODOLOGY

(Figure 5) The inputs include raw data, WiFi sensor readings, and user inputs, specifically motion data. The collected data is then analyzed using the time-splitting method, chosen for its benefits of scalability, parallel processing, granularity, resource management, incremental processing, and data exploration. The graph represents the process of setting up data, followed by analysis and the utilization of machine learning for prediction. Calculated functions and insights are derived, culminating in user actions. Ultimately, the project generates graphs and average data that users can access through their application accounts, allowing them to monitor the obtained data, such as average heartbeat, active rates derived from motion data, real time and predicted percentage of getting diseases such as COPD and asthma.

THE USE OF MOBILE APP

- The app allows users to connect their phone to the WiFi sensing device, which can provide input for a range of data that the device itself cannot obtain such as past health records and medical information . (Figure 4)
- Visualize data and make it more accessible and user-friendly by displaying the various data in a clear and easy-to-understand format, such as graphs or charts. (Figure 4)
- Real-time monitoring and remote access for caregivers to monitor users' health data, breathing patterns, activity status, etc. (Figure 4)
- The app can also provide notifications or alerts if there are any significant changes in respiratory data, which could signal a potential COPD exacerbation. (Figure 4)
- Emergency notifications will be sent to the user's caregivers or family via the app whenever fall detection or onset of respiratory symptoms happens.

COMPETITOR ANALYSIS

Comparison to other health monitoring technologies

Competitors	Description	Pros	Cons
Voice-activated assistants	Programmed to provide medication reminders, daily activity prompts, and other forms of support to help elderly. Examples are Amazon's Alexa & Google Home.	• Simplicity of voice commands • Control other smart home systems, such as lighting or temperature control.	• Not suitable for individuals with hearing or speech impairments. • Limited to provide personalized support or care.
Wearable devices	Worn on the body and can track various aspects of an individual's health, including vital signs, physical activity, and sleep patterns. Is small and unobtrusive. Examples are Apple Watch & Fitbit.	• Provide real-time data • Can be personalized	• Expensive for models that offer advanced features. • Individuals may find it uncomfortable or inconvenient to wear.
Monitoring cameras	Camera technology that allows caregivers or family members to remotely monitor the activities of an elderly individual in their home. Examples are Arlo & Nest Cam	• Remotely accessible • Real-time monitoring	• Invasion of privacy • Loss of autonomy for the individual

MARKET INSPECTION

- The global home healthcare market size is projected to reach USD 274.7 billion by 2025 from USD 181.9 billion in 2020, at a CAGR(Compound annual growth rate) of 8.6%.
- COPD impacts an estimated 251 million people worldwide. In the United States, 16 million people have COPD, and each year more than 150,000 people die because of it..
- Asthma affected an estimated 262 million people in 2019 and caused 455 000 deaths.
- Wi-Fi Sensing is supported by certain types of chipsets currently, but it will change in 2023 with 802.11bf chipsets. The number of compatible devices will increase from USD156.77M in 2022 to USD192.67M in 2023 and accelerate after 802.11bf standardization in 2024.

POTENTIAL IMPACT/OUTCOME

'Look After' will be an improved solution with the following **benefits**:

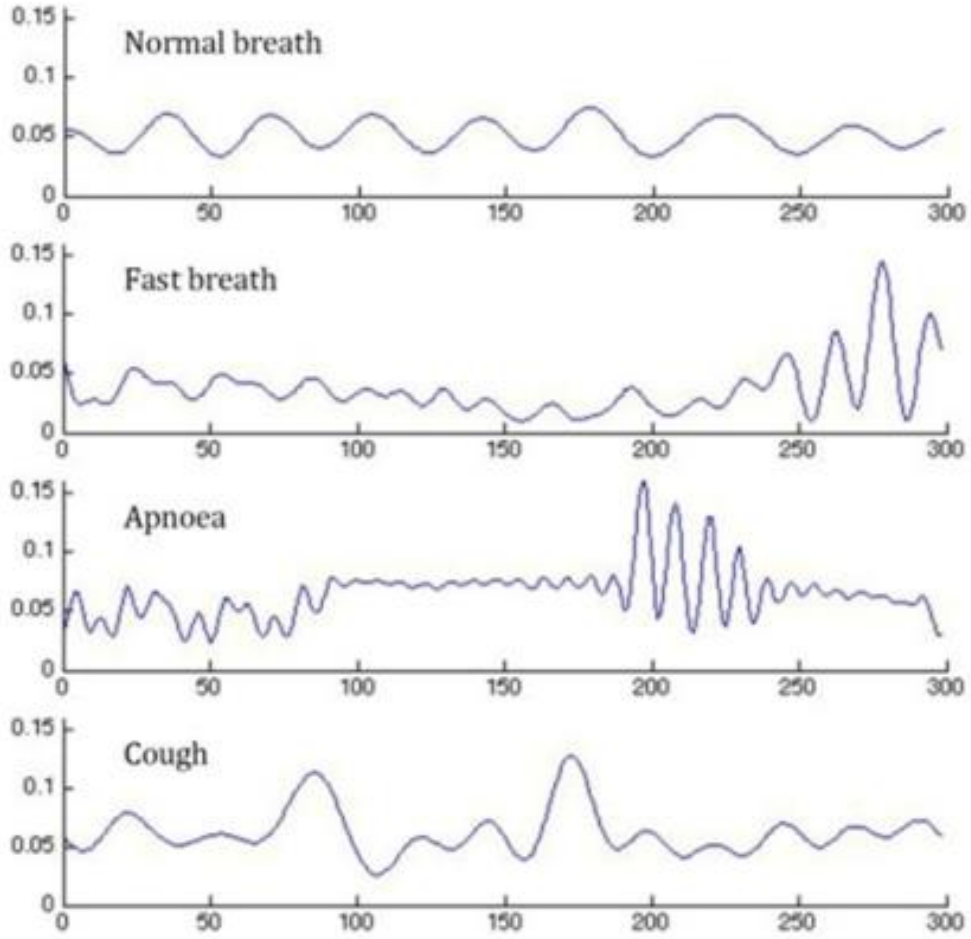
- Non-intrusive monitoring without the need for cameras or wearable devices.
- Provide real-time data on an individual's health status.
- Customized to meet individual's specific needs and preferences, providing personalized care and support.
- Relatively inexpensive as it uses existing WiFi routers to detect individual's movement and activity patterns in home.

'Look After' **target audiences and market needs** (potential outcome):

- Elderly: 'Look After' supports independent living and assists caregivers and healthcare providers in monitoring elderly individuals' well-being.
- Solitary individual: Injuries sustained from a fall can worsen and even become life-threatening but 'Look After' can detect potential falls and alert emergency services for immediate assistance to prevent fatality.
- Potential patients with respiratory diseases: COPD causes 5% of global death annually which is why detection of COPD signs for early prevention is important which 'Look After' is capable of.

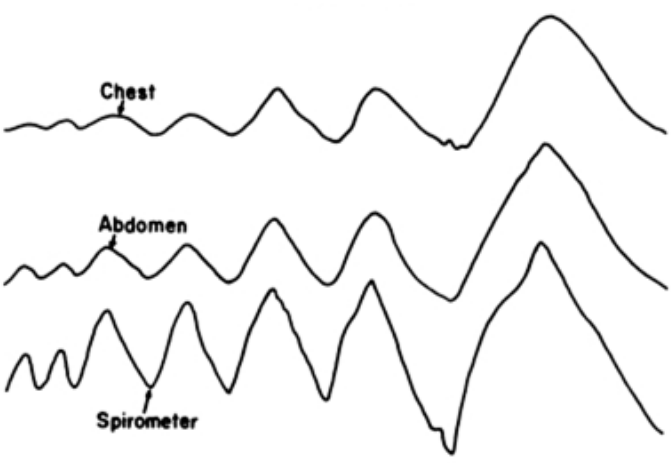
REFERENCES

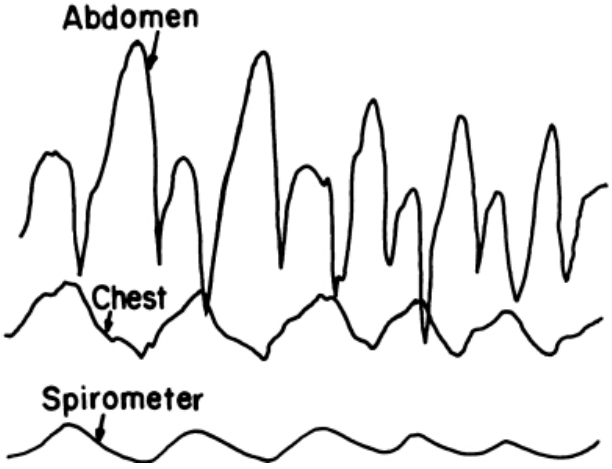
1. *MUMJ Clinical Review 23*. https://www.mangalam.nl/wp-content/uploads/2018/12/Breathing-Pattern.pdf
2. Liu M, Huang MC. Asthma Pattern Identification via Continuous Diaphragm Motion Monitoring. 2015;1(2):76-84. doi:https://doi.org/10.1109/tmscs.2015.2496214
3. Kumar Ashutosh, Gilbert RG, J. Howland Auchincloss, Peppi D. Asynchronous Breathing Movements in Patients with Chronic Obstructive Pulmonary Disease. 1975;67(5):553-557. doi:https://doi.org/10.1378/chest.67.5.553
4. Cardenas, Gutierrez, C. A., & Aguilar-Ponce, R. (2021). Influence of the antenna orientation on wifi-based fall detection systems. Sensors (Basel, Switzerland), 21(15), 5121–. https://doi.org/10.3390/s21155121
5. Ding, & Wang, Y. (2020). A WiFi-Based Smart Home Fall Detection System Using Recurrent Neural Network. IEEE Transactions on Consumer Electronics, 66(4), 308–317. https://doi.org/10.1109/TCE.2020.3021398
6. Khan, Alam Bhuiyan, M. Z., Islam, M. M., Wang, T., Zaman, A., & Tao, H. (2020). Wi-Fi Signal Analysis for Heartbeat and Metal Detection: A Comparative Study of Reliable Contactless Systems. 2019 6th International Conference on Dependable Systems and Their Applications (DSA), 81–90. https://doi.org/10.1109/DSA.2019.00018
7. C. Wu, B. Wang, O. C. Au and K. J. R. Liu, "Wi-Fi Can Do More: Toward Ubiquitous Wireless Sensing," in IEEE Communications Standards Magazine, vol. 6, no. 2, pp. 42-49, June 2022, doi: 10.1109/MCOMSTD.0001.2100111.
8. MarketsandMarkets. (2020, September 16). Global Home Healthcare Market Size to Reach $274.7 Billion by 2025. Yahoo Finance. https://finance.yahoo.com/news/global-home-healthcare-market-size-174100815.html
9. World Health Organization. (2021, May 4). Asthma. https://www.who.int/news-room/fact-sheets/detail/asthma
10. National Institutes of Health. (2018, November 14). NIH statement for World COPD Day 2018. https://www.nih.gov/news-events/news-releases/nih-statement-world-copd-day-2018#:~:text=A%20leading%20cause%20of%20death,people%20die%20because%20of%20it .
11. ABi Research. (2019). The Commercial Market Value for Wi-Fi Sensing. Retrieved from https://go.abiresearch.com/hubfs/Marketing/Whitepapers/The%20Commercial%20Market%20Value%20for%20Wi-Fi%20Sensing/ABi_Research_The_Commercial_Market_Value_for_WiFi_Sensing.pdf?hsCtaTracking=7f06d6ac-8974-4c37-91c7-2071cd1f0067%7C3bfb299f-a255-4090-be8c-9746d41d9df8



MUMJ Clinical Review 23. https://www.mangalam.nl/wp-content/uploads/2018/12/Breathing-Pattern.pdf

Figure 1: Four Respiratory Pattern of Pre-Asthma Attack Symptoms





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Figure 2: Normal Breathing Pattern

Figure 3: COPD Breathing Pattern

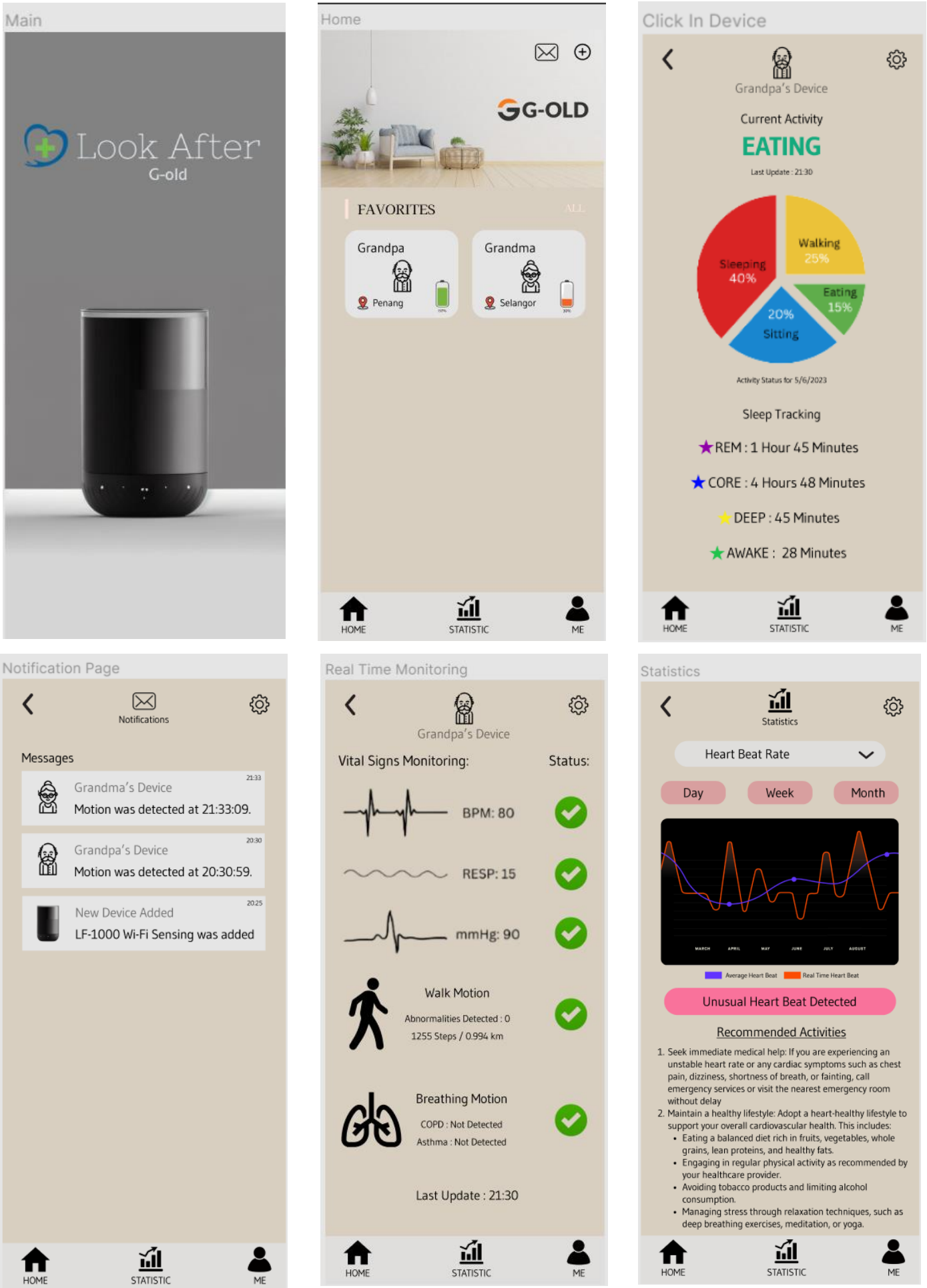


Figure 4: Proposed App Interface

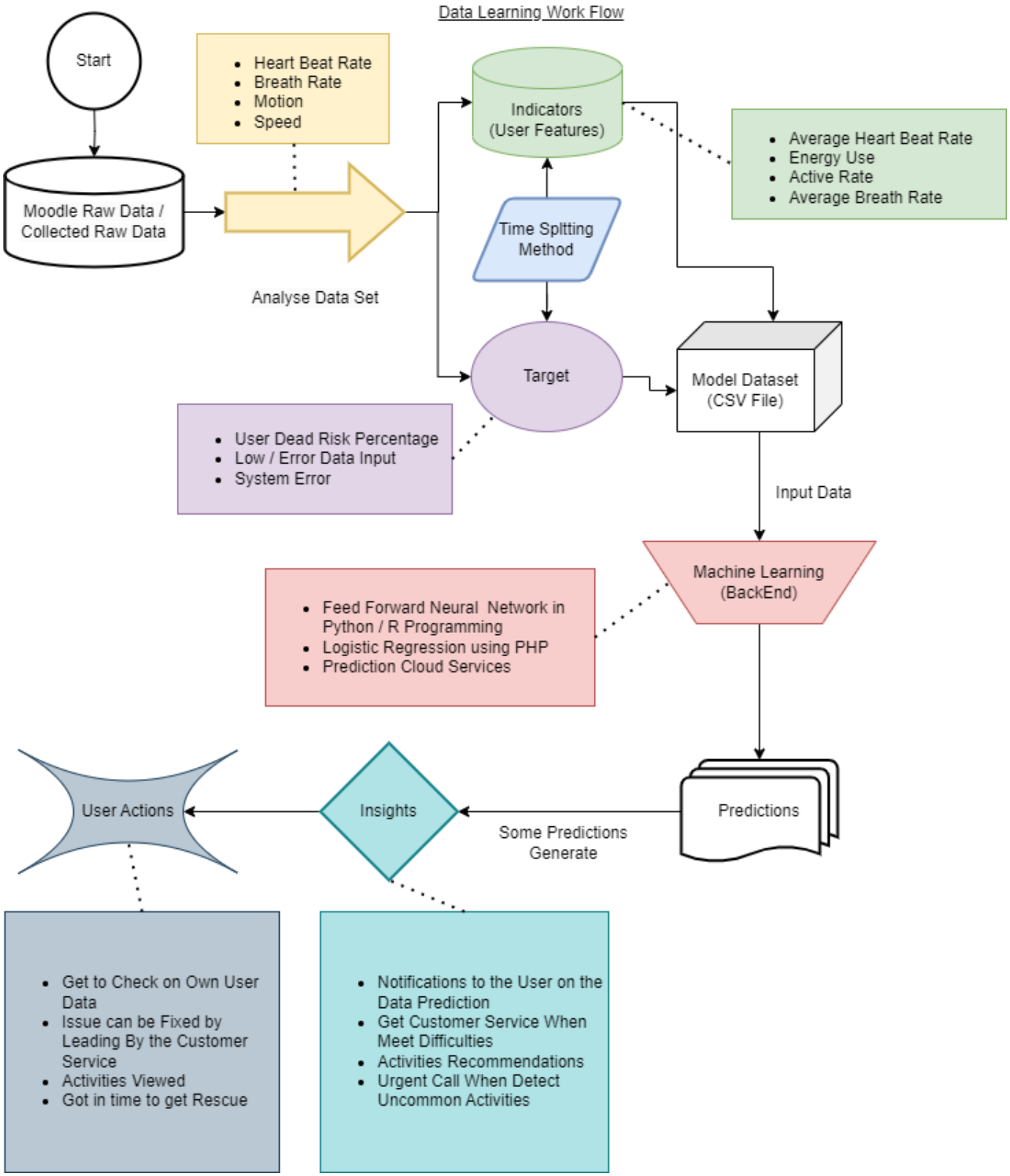


Figure 5: Data Processing Methodology